CS258

THE UNIVERSITY OF WARWICK

Second Year Winter Examinations 2016/17

Database Systems

Time allowed: 2 hours

Answer **FOUR** questions out of six.

Read carefully the instructions on the answer book and make sure that the particulars required are entered on **each** answer book.

Calculators are not required and not permitted.

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1. Consider the COMPANY schema diagram in Figure 1 overleaf.

   (a) Give an example of a database state that violates at least one of the key constraints specified by the schema. [3]

   (b) Give an example of a database state that violates at least one of the referential constraints from the schema. [3]

   (c) Write SQL code to create the following views.

      i. A view named FMAN that lists the first names of all departmental managers. [3]

      ii. A view named DEPRO that provides the following details for every department: department number, department name and the number of projects controlled by the department. [6]

      iii. A view named EMPNOT that reports the following details for each department: department number, the number of employees who do *not* work on any projects controlled by that department. [10]

   Total mark for this question: 25

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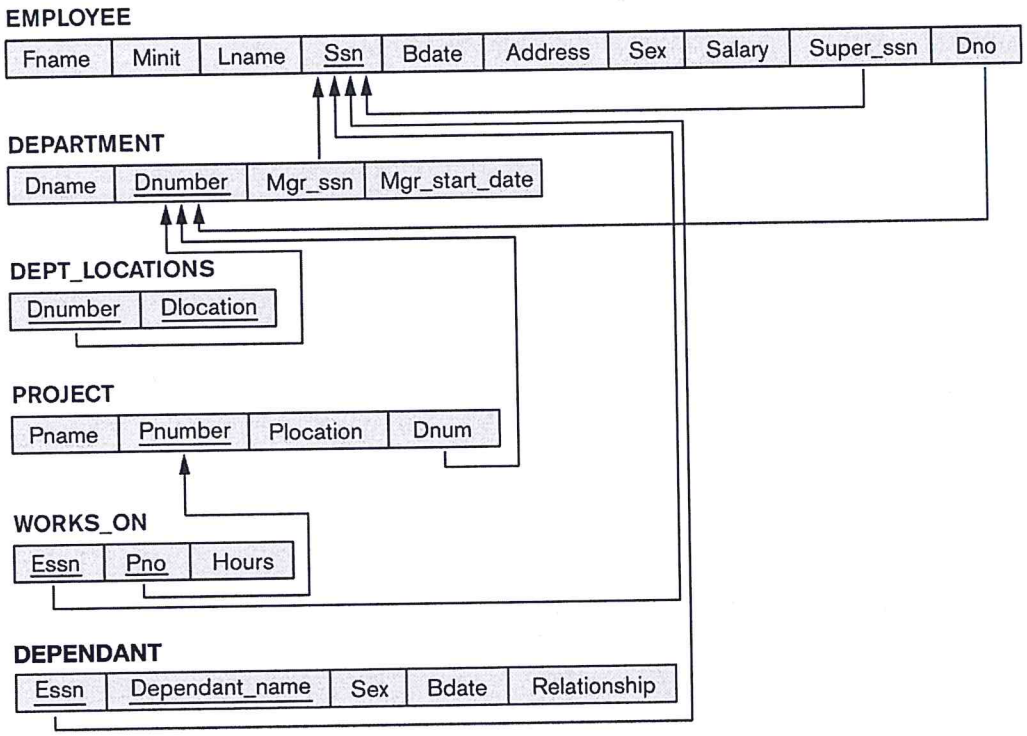


Figure 1: COMPANY schema diagram

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2. Write the following queries in SQL for the COMPANY database of Figure 1.

   (a) Retrieve the names of all dependants of the manager of the 'Education' department.[5]

   (b) List the first names of all employees who are not departmental managers. [5]

   (c) Retrieve the department numbers of all departments that employ more men than women. [5]

   (d) Retrieve the last names of managers whose departments have the largest number of locations (among all departments). [5]

   (e) For all departments, list the department number along with the number of employees (not necessarily from this department) who work on a project with a member of that department. [5]

   Total mark for this question: 25

3. Specify in relational algebra the following queries for the COMPANY database (Figure 1).

   (a) Retrieve the first names of all male employees along with their addresses. [5]

   (b) Retrieve project names of all projects that are controlled by a department with a location in Singapore. [6]

   (c) Retrieve department names of all departments whose locations include all those of Department No. 1. [7]

   (d) Print all pairs of Social Security Numbers – named *Ssn1*, *Ssn2* respectively – such that the corresponding employees do not work together on any single project. [7]

   Total mark for this question: 25

Continued

4. (a) Explain what a key is. Give an example of a superkey that is not a key. [4]

   (b) Consider the relation schema CONTESTANT($Name$, $Points$, $Town$) and the relation state given below. Identify potential superkeys of CONTESTANT.

Name	*Points*	*Town*
Peter	11	York
John	16	Bath
Adam	16	Bath
John	10	Bath
John	16	York

   [6]

   (c) Consider the relation schema $R(A, B, C)$. Suppose it is known that $\{A, B\} \rightarrow \{C\}$, $\{B, C\} \rightarrow \{A\}$ and $\{C, A\} \rightarrow \{B\}$. Discuss the shape of possible keys in R. [7]

   (d) Consider the relation schema $REL(X, Y)$. Is it possible to write a SQL query which will return an empty result if and only if the current database state satisfies $\{X\} \rightarrow \{Y\}$? [8]

Total mark for this question: 25

5. Consider the following relation schema:

$$\text{SECRETS}(A, B, C, D, E, F, G, H, I)$$

where $\{A, C, E\}$ is the primary key and each attribute domain is the set of integers. Assume the following additional dependencies:

$$\begin{array}{ll} \{A, C\} \rightarrow \{B\} & \{D\} \rightarrow \{G\} \\ \{C, E\} \rightarrow \{D\} & \{D\} \rightarrow \{H\} \\ \{A, E\} \rightarrow \{F\} & \{F\} \rightarrow \{I\} \end{array}$$

   (a) Based on the primary key, determine whether this relation is in 1NF, 2NF, 3NF. [9]

   (b) Perform any normalisation steps required to bring it into 3NF. [16]

Total mark for this question: 25

Continued

6. (a) Describe the two restrictions governing data access in the Bell-LaPadula model of security. [6]

(b) A file of 100,000 blocks is to be sorted with an available buffer space of 100 blocks. How many passes will be needed in the merge phase of the external sort-merge algorithm? [4]

(c) What is the difference between the tuple relational calculus and the domain relational calculus? Are there queries that can be expressed in the former but not the latter? Justify your answer. [7]

(d) Specify the following query in the tuple relational calculus. [8]

List the first names of employees who work on all the projects controlled by the Economics Department.

Total mark for this question: 25

End